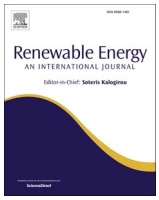
Renewable Energy 247 (2025) 123019



Contents lists available at ScienceDirect

Renewable Energy

journal homepage: www.elsevier.com/locate/renene





Future prediction on the remaining useful life of proton exchange membrane fuel using temporal fusion transformer model

Jia-Hong Chou, Fu-Kwun Wang *

Department of Industrial Management, National Taiwan University of Science and Technology, Taipei City, 106335, Taiwan

ARTICLE INFO

Keywords:
Proton exchange membrane fuel cell
Remaining useful life
Temporal fusion transformer
Future prediction

ABSTRACT

Proton exchange membrane fuel cell (PEMFC) is regarded as one of the leading developmental directions of the green power source. Predicting the remaining useful life (RUL) of online PEMFCs is challenging, primarily due to factors such as complex nonlinear degradation patterns and unknown data. To enhance the performance of RUL prediction, we present a deep learning model-based temporal fusion transformer model to predict the future total voltage of a PEMFC stack. The performance of RUL predictions was evaluated using different starting RUL prediction points to compare them with the existing study. The experiments are conducted with different future prediction steps for future RUL prediction to determine the model's prediction step limit. The proposed method achieved relative errors of 0.0148 %, 0.0060 %, 0.0662 %, and 0.0486 % for training lengths 300, 400, 500, and 600, respectively, in the FC1 and 0.0326 % in the FC2 datasets compared to the existing study scenarios. The experimental analysis resulted in relative errors of 0.0738 % for the FC1 dataset, 1.0239 % and 1.6512 % for the two FC2 dataset scenarios.

1. Introduction

The proton exchange membrane fuel cell (PEMFC) is a multi-variable, nonlinear system including multi-physics, multi-scale, and multiphase, which is challenging to model dynamically. Because of its high density, eco-friendly, and lightweight, PEMFC has become a critical power source for many transportation applications [1]. Therefore, managing PEMFC is essential by accurately forecasting the remaining useful life (RUL) prediction and making relevant optimal operational decisions. Recently, the data-driven approach based on the deep learning (DL) algorithm has been widely used for solving nonlinear and multi-characteristic modeling problems. The most popular structure is the recurrent neuron network (RNN) for predicting the RUL of PEMFC because of recurrent weight updating in recurrent cells [2]. To the best of our knowledge, making an accurate RUL prediction of dynamic PEMFC operation is challenging because of its variety of degradation patterns. The RNN and echo state network (ESN) models for single input RUL prediction developed for PEMFC static operating conditions have yet to consider the voltage recovery effect caused by the stack test start-stop [3]. The degradation of stack voltage decreases monotonically in static operating conditions. Since the voltage is readily observable in the PEMFC, it serves as a health indicator [4]. The reactant voltage, current, reactant temperature, and pressure of PEMFC contain important degradation information. Existing data-driven methods require large amounts of aging data to achieve accurate decay trend prediction for PEMFC. Using multivariate time series data, single-input prediction algorithms cannot accurately predict RUL under dynamic PEMFC operating conditions. Yue et al. [5] provided a good review on analyzing the prognostics scales, horizon, threshold, and methods such as data-driven, model-based, and hybrid methods for the RUL prediction of PEFMC. Zhang et al. [6] reviewed the new development of the state of health (SOH) prediction of PEMFC systems, including the diagnostic techniques, the health indicator (HI) approaches, and the innovative application of effectiveness of these techniques and approaches. They pointed out that these studies play a role in extending the service life of the PEMFC systems after completing the judgment of the SOH of the stack system.

Tao et al. [7] proposed a power consisting of a two-dimensional convolutional neural network (CNN2D), improved Manta-Ray foraging optimization (MRFO), and gate recurrent unit (GRU), the full hybrid DL model called CNN2D-MRFO-GRU. Using CNN2D to extract features from the nonlinear degradation characteristics and combine them with the optimal GRU model, which is optimized with an improved MRFO algorithm. The hybrid model outperformed the others compared models

* Corresponding author.
E-mail address: fukwun@mail.ntust.edu.tw (F.-K. Wang).

https://doi.org/10.1016/j.renene.2025.123019
Received 30 June 2024; Received in revised form 3 February 2025; Accepted 31 March 2025
Available online 1 April 2025
0960-1481/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

in their constructed dataset. A bi-directional long short-term memory (BiLSTM) model with multiple variables has been used to predict the RUL of PEMFC, where Gaussian filtering and cubic interpolation are used to preprocess the data [8]. A hybrid method, including the residual block, the LSTM model, and the multi-head self-attention, is proposed to predict the total voltage degradation [9]. When fuel cells are under dynamic use conditions, an attention mechanism-based data-driven framework is developed to predict their voltage degradation [10]. A hybrid method based on the convolutional neural network (CNN) and LSTM is presented to predict the voltage degradation of a fuel cell system, where the complete ensemble empirical mode decomposition is used to decompose the nonlinear and non-stationary voltage data [11]. A hybrid method based on extreme learning with an auto-encoder and fuzzy extension broad learning system is used to predict the power degradation of PEMFC [12]. A hybrid method based on CNN and BiLSTM is used to predict the RUL of PEMFC, where the Savitzky-Golay model is used to smooth the data [13].

The existing data-driven methods cannot accurately predict the RUL of PEMFC under dynamic operating conditions and require substantial training data. In addition, most studies used all data to validate their approaches. The RUL prediction deals with future forecasts. Only a few existing researches discuss the step-ahead prediction for future RUL prediction. That is, the data after the RUL prediction point are unknown. Jin et al. [14] introduced a cycle reservoir with a jump with a multi-input and multi-outputs (MIMO-CRJ) framework to enhance training efficiency and simplify the ESN model. They introduced the concepts of single-step and multi-step prediction. Multi-step predictions are attained through an iterative application of single-step predictions. It should be noted that the ESN model is not a generative model trained through sequence-to-sequence learning; hence, multi-step prediction relies on using the previously predicted value for subsequent predictions. Benaggoune et al. [15] introduced a multi-step-ahead prediction structure based on a dilated CNN with an attention mechanism for PEMFC. Their research compared the step-ahead predictions at steps five, ten, fifteen, and twenty-five with other RNN-based models. Only a few existing papers focus on predicting the future RUL for PEMFC. In addition, we should address the issue of future prediction steps. If the model can accurately predict more future steps, the operator or user can detect degradation for the next step earlier. This research aims to determine the prediction horizon limit for the deep learning model when predicting the RUL.

In recent years, generative artificial intelligence (AI) has gained popularity and is expected to be a trend in the next generation, exemplified by models like transformers and generative pre-trained transformers (GPTs) [16]. The temporal fusion transformer (TFT) model has been successfully applied in time series analysis [17]. This model employs sequence-to-sequence learning, using past input features and known future features to predict the next step in the time series data. In this research, the hour index serves as the known feature, and multiple hour indices input to perform multi-step ahead predictions. As deep learning models become increasingly complex and accurate, researchers are working on enhancing the model's interpretability to explain the contributions of each input feature, thus improving the performance of the deep learning model. The TFT model includes variable selection networks that calculate and understand feature contributions. The TFT model has the advantage of learning long-term dependencies using a self-attention mechanism; this means it can directly learn patterns during the training process, giving the model more advantage over many deep learning models in terms of robustness and stability.

This study modified the TFT model by integrating a BiLSTM encoder-decoder layer for future RUL prediction of PEMFCs. Compared to the original TFT model, the improved version demonstrates superior performance and effectively addresses the challenges of future RUL prediction at each prediction step in our experimental analysis. Our proposed approach incorporates data preprocessing methods, including the Bacon-Watts and double Bacon-Watts models [18], for defining the training set and feature selection using a random forest (RF) model. The knee-onset and knee-point identified by the Bacon-Watts and double Bacon-Watts models serve as the training and prediction starting points (Sp). To optimize the hyperparameters of the deep learning model, we employ an open-source AutoML toolkit alongside the improved TFT model in the proposed approach. During RUL prediction, our approach supports online prediction, enabling real-time forecasting when PEMFC indicators are unavailable at the time of prediction. To emphasize the issues addressed in this paper, our proposed approach for online RUL prediction differs from traditional SOH estimation, which assumes that input variables are fully known during prediction. In summary, the proposed approach motivated by some limitations of DL models for online RUL prediction concerning long-term and local dependency learning ability, future prediction steps, and the nonlinearity relationship between the PEMFC HI and input features, a deep learning model-based improved TFT is developed with tree-structure Parzen estimator (TPE) involved in an open-source AutoML for hyperparameter tuning and quantile prediction interval for uncertainty estimation. The main contributions and novelties of this paper are:

1. The original TFT model's LSTM encoder component was replaced with a BiLSTM encoder to capture and extract both forward and backward sequence features.
2. The Bacon-Watts and double Bacon-Watts models define the training set's significant battery degradation trend to improve the DL model's performance on future RUL prediction problems.
3. A detailed discussion of feature selection using an RF model based on feature domain visualization helps the reader understand its importance.
4. This research employs an open-source AutoML toolkit to fine-tune the hyperparameters using the TPE algorithm and visualize the convergence of hyperparameter optimization for the improved TFT model.
5. Different prediction steps of the improved TFT model for future RUL prediction are demonstrated to determine its prediction limits and comprehensively analyze its predictive behavior in the experimental analysis.

The overall organization of the paper is as follows. Section 2 presents the dataset and RUL definition. Section 3 describes the proposed method and its implementation. Section 4 provides the experimental validation for the proposed improved TFT prediction method and further results analysis. The conclusion is presented in Section 5.

2. Dataset description

2.1. Row dataset introduction

To verify the performance of the proposed approach in PEMFC online RUL prediction, we selected one of the most widely used PEMFC degradation datasets from the IEEE PHM 2014 Data Challenge [19] for experimental analysis. FC1 and FC2, which consist of two long-term PEMFC stack experiments with 5-cell stacks and an active area of 100 cm^2, were chosen to validate the proposed approach. FC1 was tested under stationary conditions in a current of 70A, and each characterization was measured every week at times 0, 48,185, 348, 515, 658, 823, and 991 h. FC2 was operated under a dynamic ripple current of 70A with 7A oscillations at a frequency of 5 kHz, and the characterization was estimated every week at 0, 35, 182, 343, 515, 666, 830, and 1016 h. Twenty-five characterizations for FC1 and FC2 datasets to monitor the battery's status and prognostic during each measured time interval are shown in Table 1. Compared to other prognostic applications, publicly available datasets for PEMFC prognostics are limited. Therefore, we focus on FC1 and FC2 data from the IEEE PHM 2014 Data Challenge, the only publicly available open-source dataset. To enhance dataset diversity, we designed three different experimental scenarios, each

Table 1
The characterizations of the FC1 and FC2 datasets.

Index	Characterization (Feature)	Description
1	Time	Aging time (hour)
2	V_1	Cell#1 voltage (V)
3	V_2	Cell#2 voltage (V)
4	V_3	Cell#3 voltage (V)
5	V_4	Cell#4 voltage (V)
6	V_5	Cell#5 voltage (V)
7	V_{total}	Stack voltage (V)
8	I	Current (A)
9	J	Current density (A/cm^2)
10	TinH2	The inlet temperatures of hydrogen (^{o}C)
11	TourH2	The temperatures of hydrogen (^{o}C)
12	TinAIR	The inlet temperatures of air (^{o}C)
13	ToutAIR	The outlet temperatures of air (^{o}C)
14	TinWAT	The inlet temperatures of cooling water (^{o}C)
15	ToutWAT	The outlet temperatures of cooling water (^{o}C)
16	PinAIR	The inlet pressure of air (mbara)
17	PoutAIR	The outlet pressure of air (mbara)
18	PinH2	The inlet pressure of hydrogen (mbara)
19	PoutH2	The outlet pressure of hydrogen (mbara)
20	DinH2	The inlet flow rate of hydrogen (l/mn)
21	DoutH2	The outlet flow rate of hydrogen (l/mn)
22	DinAIR	The inlet flow rate of air (l/mn)
23	DoutAIR	The outlet flow rate of air (l/mn)
24	DWAT	The flow rate of cooling water (l/mn)
25	HrAIRFC	The inlet hygrometry (Air) estimated (%)

incorporating five different prediction steps. Additionally, we included comparisons based on scenarios from previous studies to evaluate the proposed approach comprehensively.

Understanding which characteristics best represent battery degradation is crucial before conducting experimental analysis. Based on previous studies, several researchers have emphasized that the most essential and significant characteristic in the FC1 and FC2 datasets is the degradation of PEMFC total voltage [15]. The total voltage is the sum of the voltages of all individual cells connected in series within the cell pack. The feature of V_{total} represents the total voltage in both datasets, which performs the polarization of stacked cell degradation. Most studies analyze V_{total} as the response variable for PEMFC RUL prediction [16]. Thus, in this study, the characterization of V_{total} is considered the response variable, while the other features serve as independent variables in the proposed approach.

2.2. PEMFC end of life determination

Determining a suitable end of life (EOL) threshold is essential in the RUL prediction. In the real world, the EOL determination is based on the application. If the application requires the PEMFC charge-discharge low frequency, the EOL threshold would be defined as a high value to renew the PEMFC when its degradation to the threshold early. This paper investigates the related papers using the same dataset with us for the RUL prediction. The investigation results for EOL determination are shown in Table 2. Based on Table 2 EOL threshold reference, this study chooses the EOL threshold as 96 % of the initial V_{total}. However, the EOL may differ even if the same threshold is selected because of different data preprocessing or personal determination. In Fig. 1, two points are lower than the chosen threshold of the FC1 total voltage degradation curve. This paper assumes that the first point drop to 96 % is the EOL of FC. Finally, the EOL is determined as 815 h for the FC1 dataset, which is reached to other research and can compare the RUL prediction results.

This research uses the same EOL threshold as 96 % of the initial V_{total} to identify the EOL hours for FC2. However, determining the EOL hour can be challenging due to the variability of the V_{total} degradation curve, as demonstrated in Fig. 1. Table 3 shows two different EOL hour determination criteria based on references for the FC2 dataset. In the first scenario, some studies select the EOL hours to exceed 500, and the

Table 2
Training length and EOL threshold comparison on IEEE PHM 2014 Challenge dataset.

Ref.	Training length	EOL threshold
[8]	FC1: 0 h- 300, 400, 500, 600 h FC2: 0 h- 300, 400, 500, 600 h	96 % of the initial V_{total} (FC1 is 811h and FC2 is 394h)
[12]	FC1: 0 h-340, 350, 600, 700 h	96 % of the initial V_{total}
[15]	FC1: 0 h- 591, 691, 791, 891, 991 h FC2: 0 h- 681, 731, 781, 831, 881, 931h	96 % of the initial V_{total} (FC1 is 1028h and FC2 is 958h)
[18]	FC1: 0 h- 576h FC2: 0 h- 509h	96 % of the initial V_{total} (FC1 is 927h and FC2 is 837h)
[20]	FC1: 0 h- 200, 300, 400, 500 h FC2: 0 h–200 h	96.5 % of the initial V_{total} (FC1 is 809h and FC2 is 395h)
[21]	FC1: 0 h–550 h	96.5 %, 96 %, 95.5 %, 95 %, and 94.5 % of the initial V_{total}
[22]	FC1: 0 h-500, 550, 600, 650, 700, 750 h FC2: 0 h-510, 560, 610, 660, 710, 760 h	96.5 %, and 95.5 % of the initial V_{total} for FC1 and FC2 (FC1 is 792h and FC2 is 758h)

corresponding training length is more than 500. Conversely, in the second scenario, other studies set the EOL hour to be less than 500, with training durations less than 500 h.

Following their respective logic, this research establishes two scenarios: if the EOL hours index exceeds 500 h, the training duration should exceed 500 h; otherwise, it should be less than 500 h. In this study, experiments utilize the first and last data points, representing a 96 % capacity drop, for EOL times of 393 and 878 h for FC2 under both scenarios. These EOL hours were selected because they are consistent with prior research, allowing for a comparative analysis of RUL prediction performance.

3. Proposed method

This research determines the EOL in advance to define the failure hours and build the experimental studies for FC1 and FC2 datasets. Fig. 2 depicts the flowchart of the proposed method for the online RUL prediction for the PEMFC dataset.

3.1. Raw data analysis

After getting the EOL point by the EOL threshold, the Bacon-Watts model and the double Bacon-Watts model calculate the knee-point and knee-onset of the V_{total} curve to determine the training length of offline model training. The training set is the period between knee point and onset. The reason for choosing the Bacon-Watts model and the double Bacon-Watts model to define the training set for the battery online RUL prediction problem is that the knee-point identified by the Bacon-Watts model indicates when the degradation trend begins to decline sharply. In contrast, the knee-onset point detected by the double Bacon-Watts model provides an early warning before the degradation trend accelerates. These two models help define the training set by removing redundant stable trends, reducing data interference, and enhancing model performance for online RUL prediction of PEMFC degradation aging. This period effectively represents the accelerated PEMFC degradation process due to the mathematical definitions of the knee-point and knee-onset (see the following equations).

Equation (1) represents the Bacon-Watts model for knee-point estimation, and Equation (2) is the double Bacon-Watts model, which is modified from the Bacon-Watts model for estimating the knee-onset.

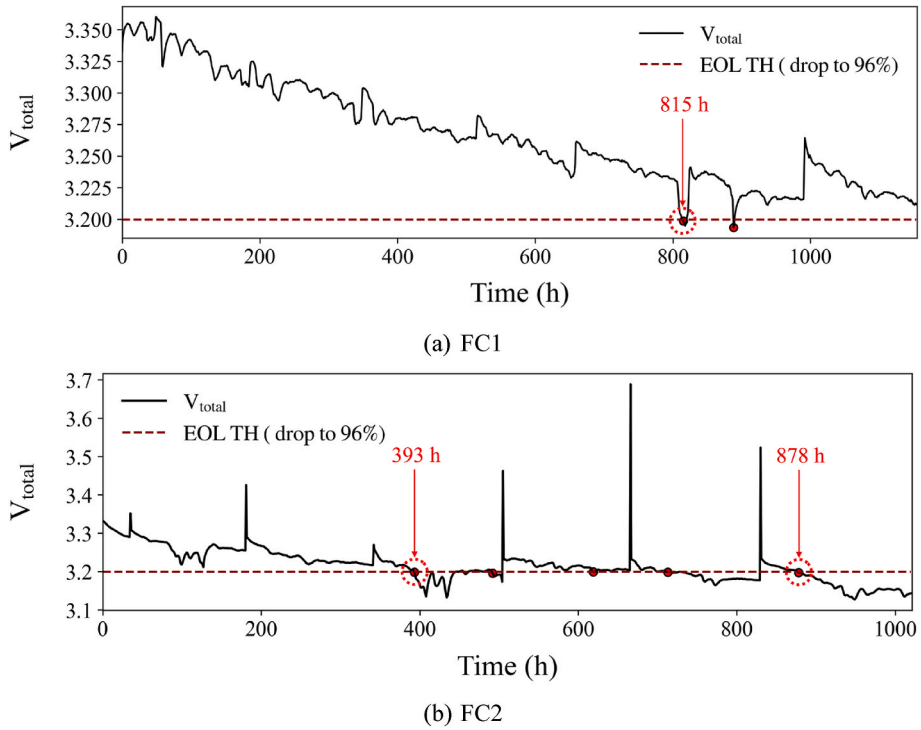


Fig. 1. The EOL hour determination for the FC1 and FC2 datasets.

Table 3
Summary of EOL, training length, and Sp for the experimental scenarios (unit = hour).

Dataset	EOL	Training length	Sp
FC1	815	116–578	578
FC2_scenario#1	393	147–186	186
FC2_scenario#2	878	568–609	609

The concept of the Bacon-Watts model is that using an unknown x_1 transition point to separate the line into two straight lines, which represent the left and right parts, and the hyperbolic tangent (tanh) function uses to give a ratio as slope to let two straight lines envelop the original degradation curve. The double Bacon-Watts model adds one more transition point x_2 for calculating one more straight line to envelop the original degradation curve. The transition point of the Bacon-Watts model represents the knee-point, which means the aging trend degrades sharply after this point, and the first transition point of the double Bacon-Watts model represents the knee-onset, which means the aging trend starts to degrade after this point [13].

$$\widehat{V}_{total} = \alpha_0 + \alpha_1(x - x_1) + \alpha_2(x - x_1)tan\,h[(x - x_1)/\gamma] + Z, \quad (1)$$

$$\widehat{V}_{total} = \alpha_0 + \alpha_1(x - x_1) + \alpha_2(x - x_1)tan\,h[(x - x_1)/\gamma] + \alpha_2(x - x_2)tan\,h[(x - x_2)/\gamma] + Z, \quad (2)$$

where α_0, α_1, and α_2 are the coefficients of each straight line, x_1 and x_2 are the transition points to separate the straight line, γ is a constant small value, Z is the random bias from Gaussian distribution $N(0,1)$, and x and $\widehat{V}_{total}$ are the independent variable and dependent variable, respectively.

However, the raw data of FC1 and FC2 have some noise and significant peaks because of the polarization and the Electrochemical Impedance Spectroscopy (EIS) measurement and should be smoothed and filtered. The high peak value and noise may affect the knee-point and knee-onset identification. This research filters both the peaks for the FC1 and FC2 datasets. The significant peaks are replaced using the interpolation method, and noise values are smoothed by locally weighted regression. The replaced value calculates the interpolation based on the next one and two polarization estimation peaks, see Equation (3).

$$X_t = \left(\frac{X_{t+2} - X_{t+1}}{H_{t+2} - H_{t+1}}\right)(H_t - H_{t+1}) + X_{t+1} \quad (3)$$

where X is the variable for removing the high peak, H is the variable of hours, and t is the time index.

This research uses not only the method for the V_{total} degradation curve but also all features of the same time index at the polarization time 0, 48, 185, 348, 515, 658, 823, 991 h for the FC1 dataset and 0, 35, 182, 343, 515, 666, 830, and 1016 h for the FC2 dataset (see Fig. 3). The results from the FC2 dataset show that the interpolation method can filter high peak and replace the value with the linear relation with the following one and two values.

The locally weighted regression model is selected for smoothing the noise data for knee-point and knee-onset identification to determine the training length and Sp hours. The benefit is that the data numbers do not reduce after smoothing. Because our EOL determination for FC2 assumes two EOL hours in Section 3.1, here this study identifies knee-point and knee-onset for three different scenarios for our experiments, which are FC1, FC2 senario#1 (EOL = 393), and FC2 senario#2 (EOL = 878). The following shows the formulation of locally weighted regression. To get the local weights and linear regression coefficients, this research must

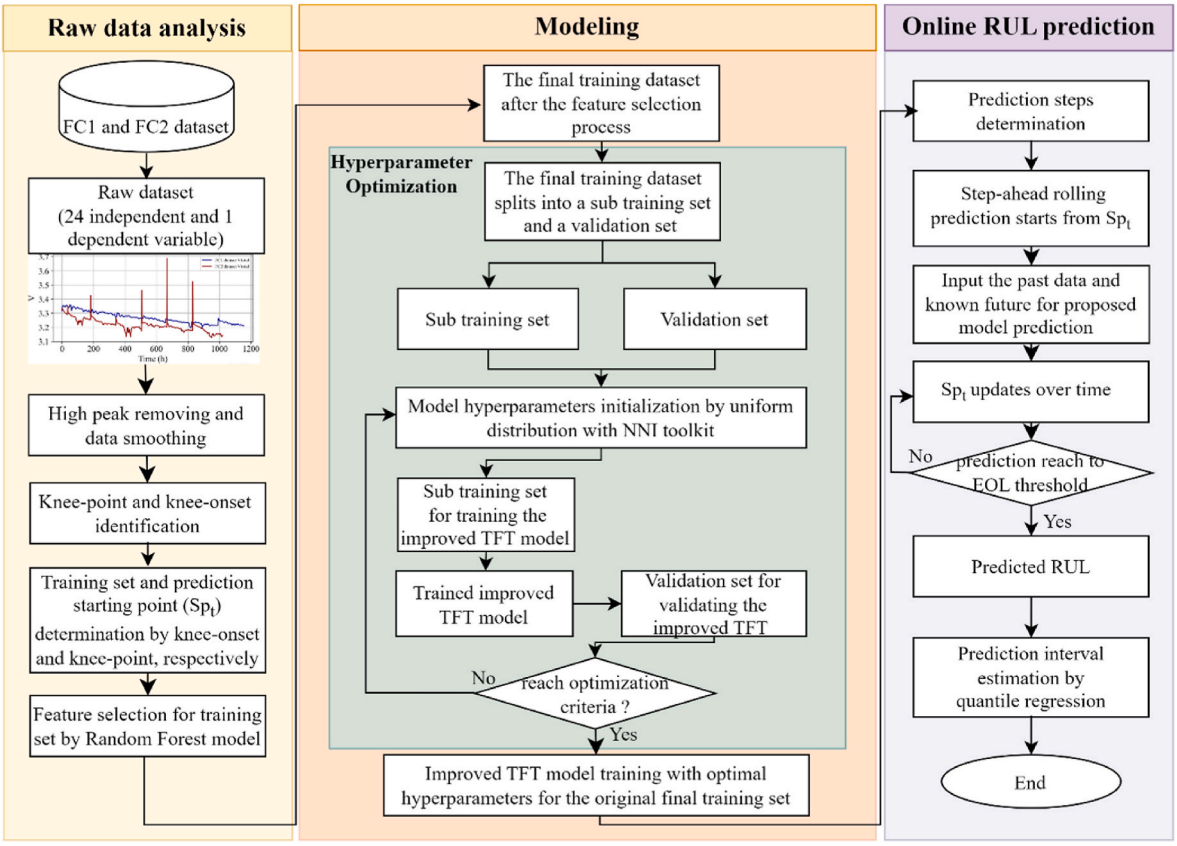
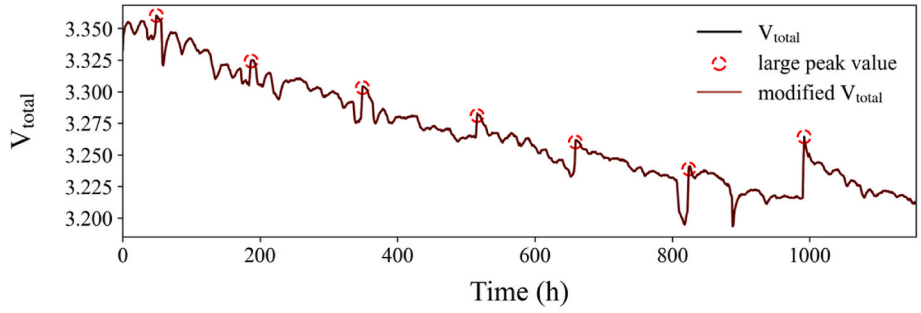
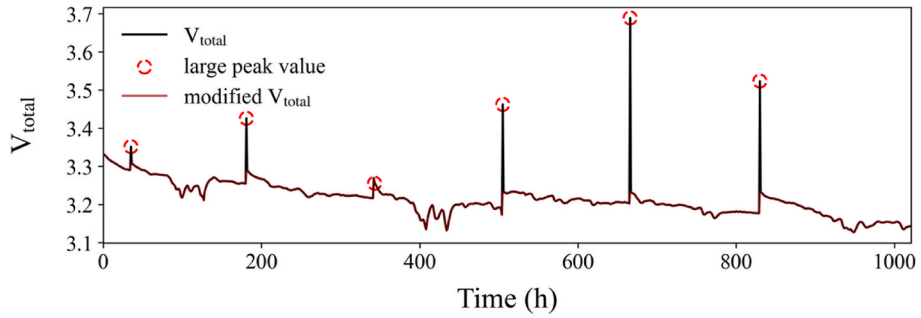


Fig. 2. Flowchart of the proposed method.



(a) FC1



(b) FC2

Fig. 3. The modified V_{total} degradation curve for the FC1 and FC2 datasets.

minimize the cost function to estimate the smoothed transformation V_{total} degradation data. The smoothed V_{total} degradation data based on the parameters of the local weight and linear regression coefficient is given in Equation (6).

$$minimize: \sum_{i=1}^{m} w^{(i)} \left(\theta^T x^{(i)} - y\right) \tag{4}$$

$$w^{(i)} = exp\left(\frac{-(x^{(i)} - x)^2}{2\tau^2}\right) \tag{5}$$

$$\hat{y}_{smoothed} = w^{(i)}\ \theta^T x^{(i)} + \xi \tag{6}$$

The $w^{(i)}$ represents the local weight based on number i data point, θ is the linear coefficient vector, τ is the bandwidth parameter that controls the critical relationship between $x^{(i)}$ and x. Suppose the parameter of τ reach 1. The $\hat{y}_{smoothed}$ becomes more smoothed; Otherwise, it becomes more like the original y. The ξ represents the bias vector from Gaussian distribution, and x and y are represent the independent and dependent variables, respectively.

Fig. 4 shows the results of three scenarios' smoothed V_{totol} degradation curve and knee-point and knee-onset identification for determining the training set and Sp. The EOL, training length, and Sp for the FC1 dataset, FC2 scenario#1, and FC2 scenario#2 are 815 h, 116–578 h, and 578 h, and 393 h, 147–186 h, and 186 h, and 878 h, 568–609 h, and 690 h. The summary of three scenario's EOL, training length, and Sp are shown in Table 3.

There are 25 characterizations in the FC1 and FC2 datasets. This dataset's response variable, V_{total}, is derived from the summation of V_1, V_2, V_3, V_4, and V_5. The time variable, represented by the time index, has been excluded from feature selection due to its non-variable nature. Therefore, the total independent variables for feature selection are 18 characterizations. However, some characterization or so-called features are not essential and have no relationship with the V_{total} variable. Although deep learning models can select the most crucial feature during training, finding the most critical feature before model training is essential. Feature selection can reduce the data dimensionality and model training consumption time and denoise the dataset to increase the model performance. The RF model selects the essential features and, based on the distribution domain of the selected feature, explains the importance of feature selection in this study. Selecting the data length from 0 h to prediction Sp for feature selection in the FC1 and FC2 datasets. By calculating each normalized feature's importance from high to low, this research ultimately selects features whose cumulative importance score surpasses 80 %. The RF model efficiently calculates the important feature score based on the decrease in mean impurity (MDI). The RF model ensemble the feature important from its based Decision Tree (DT) model, and it can reduce the variance of important scores and overfitting problems between each based model. In the regression problem, the RF model utilizes mean squared error (MSE) as the objective function for fitting the optimal weight. Thus, the formula for the feature important score includes MSE loss. The important feature formulation and computing step is shown in the description below.

Step 1: Calculate the information gain for the parent node with the MSE of node and node weight.

$$IG_{p_j} = W_{p_j} MSE_{p_j} - \sum_{i=0}^{n} W_{c_i} MSE_{c_i} \tag{7}$$

Where p_j represents the parent node j and c_i represents the child node i. Each node's weight is W, and IG represents the information gain value.

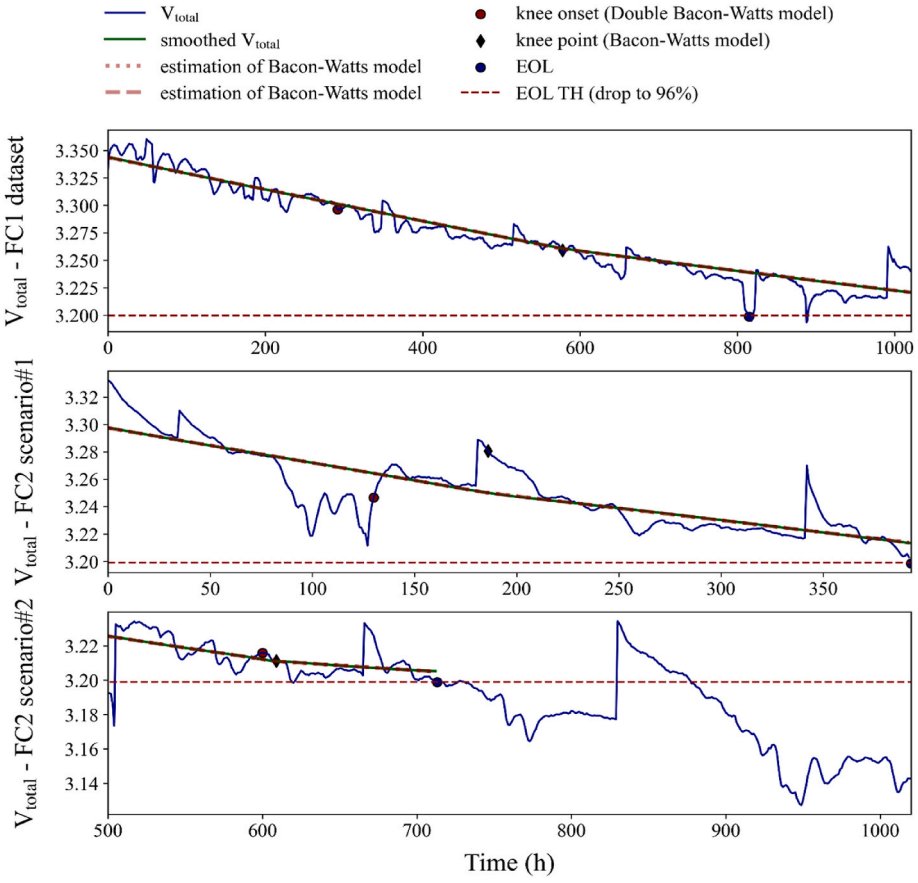


Fig. 4. The knee-onset and knee-point identification results for the FC1, FC2_scenario#1, and FC2_scenario#2.

The variable n represents the total number of child nodes.

Step 2: Compute the feature importance for each feature in a DT model

$$Important(f_k)_{DT} = \frac{\sum_{d \in all\ nodes\ branched\ by\ feature\ k\ in\ a\ DT\ model} IG_{f_{kd}}}{\sum_{m \in all\ nodes\ in\ a\ DT\ model} IG_m} \quad (8)$$

Where f_k represents the feature k. $IG_{f_{kd}}$ is the value of information gain in node d of feature k. The IG_m represents the information gain of each node in a DT model.

Step 3: Ensemble the feature important score to get the feature important score for the RF model by taking the average from each DT model's normalized feature important score.

$$Important(f_k)_{DT\ norm} = \frac{Important(f_k)_{DT}}{\sum_{k}^{v} Important(f_k)} \quad (9)$$

$$Important(f_k)_{RF} = \frac{\sum_{t \in all\ DT\ models} Important(f_t)_{DT\ norm}}{N_T} \quad (10)$$

Where $Important(f_k)_{DT\ norm}$ is the normalized feature important score of features k for a single DT model and the variable v represents the total number of features of the input data. The N_T and t represent the total number of DT models of a RF model.

Fig. 5 shows the cumulative normalized feature score from high to low for the FC1, FC2_scenario#1, and FC2_scenario#2 datasets. The selected features for FC1, FC2 scenario#1, and FC2 scenario#2 are shown in Table 4.

This research uses principal components analysis (PCA) to compose the dimensionality of the features into two dimensions and visualize the data distribution by PC1 and PC2. The eigenvalue and eigenvector are calculated by data length from 0 h to knee-point, which is prediction Sp the feature distribution domain analysis using PCA shown below.

Step 1: Find the eigenvalues and eigenvectors.

$$Ax = \lambda x \quad (11)$$

where x, and λ is the eigenvectors and eigenvalues of A, and A is the standard normalized features.

Step 2: Select the first top largest two eigenvalues and project the standard normalized features to the corresponding eigenvectors.

$$\hat{x} = \Gamma A \quad (12)$$

Table 4
The selected features for FC1, FC2_scenario#1, and FC2_scenario#2.

Dataset	Selected features	Importance score
FC1	DoutAIR (l/mn)	0.6960
	TinWAT	0.0696
	DoutH2 (l/mn)	0.0548
FC2_scenario#1	DoutAIR (l/mn)	0.3880
	TinWAT	0.2707
	J (A/cm^2)	0.0624
	I (A)	0.0582
	ToutAIR	0.0451
FC2_scenario#2	J (A/cm^2)	0.3331
	I (A)	0.2718
	TinWAT	0.2473

where $\hat{x}$, and Γ is the dimensionality reduction features vector and the first top largest two eigenvalues corresponding truncated eigenvectors of A, and A is the standard normalized features.

Step 3: Visualize the first and second dimensionality reduction feature by scatter plot and see the transformed data distribution.

Fig. 6 employs PCA to analyze the distribution across these three datasets. Feature selection diminishes noise and streamlines the data distribution. It can improve the data-driven model to speed up the training: consumption time and model performance. Comparing the two reduced dimensions of data distribution with and without feature selection reveals that PCA with feature selection yields superior results for the three datasets.

3.2. Improved TFT model

After analyzing raw data, critical features are selected, and the training set for the proposed improved TFT model is established. The improved TFT model modifies from the TFT model, an attention-based constriction model with the RNN-based variable selection, decoder and encoder structure, and the gated residual network (see Fig. 7).

3.2.1. Variable selection network

The variable selection network consists of linear transformation layers for continuous features, gated unit network (GRN) to select essential variables automatedly during the model training.

In the improved TFT model, the variable selection network consists of two important components. The first component combines with linear transformation layers to increase the dimensionality by univariate feature at time t, and the output of each linear transformation layer operates non-linear processing by its own GRN to share the weight

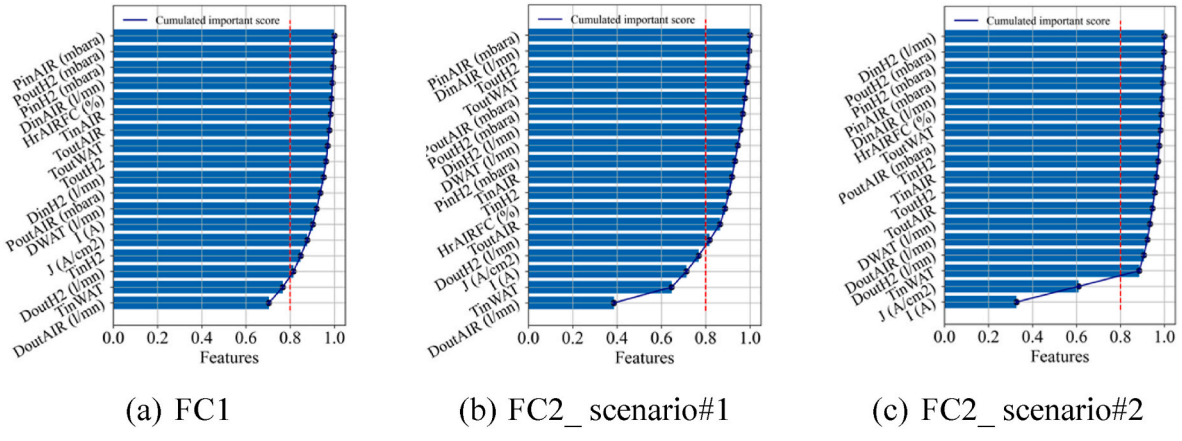


Fig. 5. The cumulated important score for the FC1, FC2_scenario#1, and FC2_scenario#2 datasets.

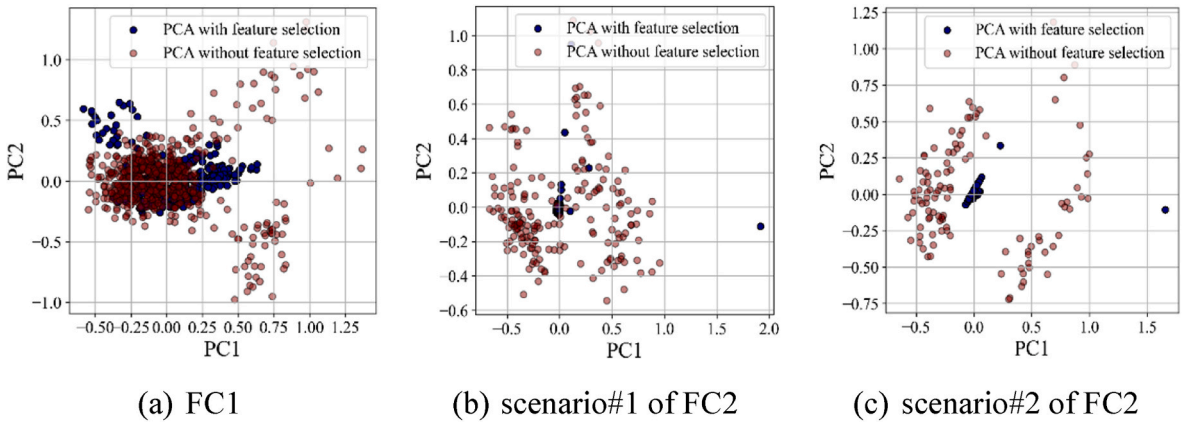


Fig. 6. The difference in distribution between with and without feature selection for each dataset.

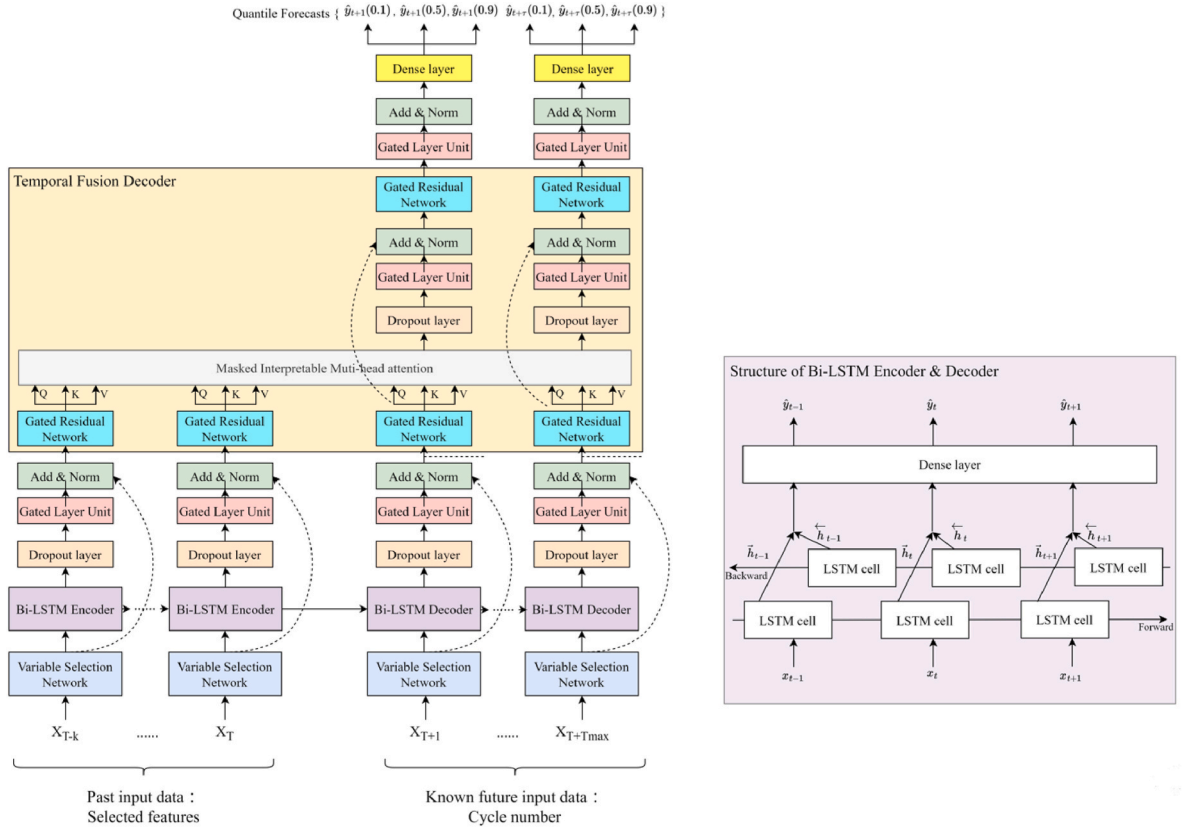


Fig. 7. The full improved TFT model structure.

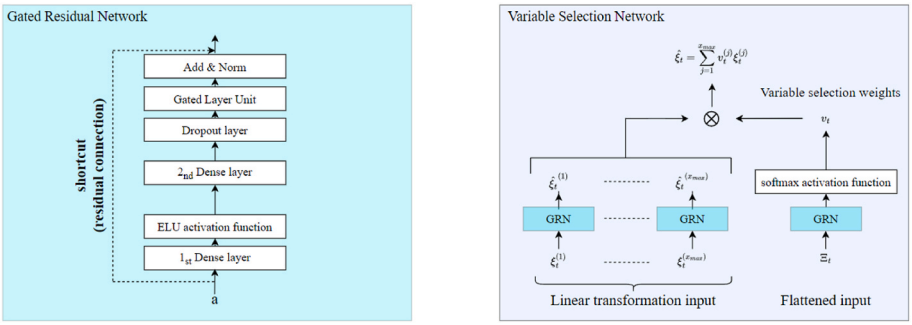


Fig. 8. The constriction of GRN and variable selection network.

across time t and prevent the gradient vanish problem. The first component operation, and GRN shown in following. Fig. 8 shows the model construction of GRN.

$$\widehat{\xi}_t^{(i)} = GRN_{\xi_t^{(i)}}\left(\xi_t^{(i)}\right), i = 1, 2, \ldots, X_{max} \tag{13}$$

where $\xi_t^{(i)}$ represent the linear transformation output of i variable at time t, and $\widehat{\xi}_t^{(i)}$ is the output of each GRN for i variable at time t. The feature numbers present as X_{max}.

$$GRN(a) = LayerNorm(a, GLU(\eta_2)) \tag{14}$$

$$\eta_1 = ELU(W_1 a + b_1) \tag{15}$$

$$\eta_2 = W_2\eta_1 + b_2 \tag{16}$$

where W_1 and W_2 represent the first and second dense layer weights, b_1 and b_2 is the bias of the first and second layer, the η_1 and η_2 is the output of the first and second dense layer., and the gated linear united (GLU) is the filter gate see Equation (17).

$$GLU\ (\gamma) = sigmoid(W_3\gamma + b_3)\bigotimes(W_4\gamma + b_4) \tag{17}$$

where γ is the input of the GLU, W_3, W_4, b_3, b_4 are the weights and bias for computing GLU, and $\bigotimes$ is the point-point multiplication. The input vector of GRN, exponential linear unit (ELU), and LayerNorm represent as a, the standard layer normalization, and the ELU activation function.

The second component of the variable selection network computes the variable selection weights by calculating the softmax activation of the flattened inputs. The flattened input is the whole features at time t which can represent as $\Xi_t = \left[\xi_t^{(1)^T}, \xi_t^{(2)^T}, \ldots, \xi_t^{(x_{max})^T}\right]^T$. The second components of variable selection network operation shown in below equation.

$$v_t = softmax\ (GRN(\Xi_t)) \tag{18}$$

where v_t is the weight of variable selection at time t of whole features, Ξ_t is the whole features at time t, and $softmax$ is the softmax activation function.

Finally, the first and second component combine as the output of variable selection network see following equation.

$$\widehat{\xi}_t = \sum_{j=0}^{X_{max}} v_t^{(i)}\xi_t^{(i)}, i = 1, 2, \ldots, X_{max} \tag{19}$$

where $v_t^{(i)}$ is the weight of variable selection at time t of number i columns, $\widehat{\xi}_t$ is the output of variable selection network combines the first and second component at time t.

3.2.2. Encoder and decoder construction for locally enhancement

The original decoder and encoder are the LSTM model. It consists of three gates: input, forget, and output gate to control the weights, called memory updating of hidden state and cell state. In the study, the LSTM encoder and decoder structure are substituted with BiLSTM to augment the hidden and cell state memory capture. The BiLSTM model considers the forward and backward direction sequence relationship to update hidden state and cell state weights. The following shows three gates: hidden state and cell state updating formulation. The forget gate F_t of the LSTM model controls the percentage of the previous hidden state h_{t-1} and current input x_t by a sigmoid function. If the output of the forget gate reaches 1, the relationship between the previously hidden state and current input information is very high. It should be kept and point-by-point multiply to the cell state C_{t-1} to affect the recurrent update. The input gate I_t updates the cell state C_{t-1} by two essential parts. The first part computes the previously hidden state h_{t-1} and current input x_t by a sigmoid function representing the value informational computation, and the second part calculates the previous hidden state h_{t-1} and current input x_t by a hyperbolic tangent (tahn) function representing the value normalize to −1 to 1. These two parts take point-by-point multiplication and add to the cell state C_t to update the weights. The output gate controls the information updating of the hidden state weights by a sigmoid function. Finally, the previous cell state C_{t-1} takes tanh function and multiplies the results of the output gate for updating the hidden state H_t.

$$F_t = sigmoid\left(W_f x_t + H_f h_{t-1} + b_f\right). \tag{20}$$

$$I_t = sigmoid(W_i x_t + H_i h_{t-1} + b_i) \otimes tahn(W_c x_t + H_c h_{t-1} + b_c). \tag{21}$$

$$C_t = F_t \otimes C_{t-1} + I_t. \tag{22}$$

$$O_t = sigmoid(W_o x_t + H_o h_{t-1} + b_o) \tag{23}$$

$$h_t = O_t \otimes tahn(C_{t-1}) \tag{24}$$

where the F_t, I_t, C_t, O_t, and h_t represent the output of the forget gate, input gate, cell state, output gate, and hidden state at number t LSTM cell, respectively. W_f, W_i, W_o, and W_c represent the weights of forget gate, input gate, cell state, output gate, and hidden state, respectively. The H_f, H_i, and H_o are the weight parameters of the hidden state of the forget gate, input gate, and output gate, respectively. The biases of the forget gate, input gate, cell state, and output gate are represented as b_f, b_i, and b_o, respectively. The point-point multiplication means the sign of $\otimes$.

The encoder and decoder structure of the improved TFT model replaces the Bi-LSTM with LSTM. The Bi-LSTM consists of two LSTM layers, and the hidden state connects forward and backward in the first LSTM layers $LSTM_{forward}$ and second LSTM layers $LSTM_{backward}$, respectively. Finally, the output of the Bi-LSTM decoder and decoder is the addition of the forward $\vec{h}_t$, the backward $\overleftarrow{h}_t$ hidden state, and the bias vector $b_{\hat{y}_t}$.

$$\vec{h}_t = LSTM_{forward}\left(x_t, \vec{h}_{tt-1}\right) \tag{25}$$

$$\overleftarrow{h}_t = LSTM_{backward}\left(x_t, \overleftarrow{h}_{t-1}\right) \tag{26}$$

$$\hat{y}_t = W_{forward}\vec{h}_t + W_{backward}\overleftarrow{h}_t + b_{\hat{y}_t} \tag{27}$$

where $W_{forward}$ and $W_{backward}$ are the BiLSTM output layer weights of the forward and backward LSTM layers, and the $\overleftarrow{h}_t$ and $\vec{h}_t$ represent the hidden state of the forward and backward LSTM layers. The input, output, and output layer biases of the BiLSTM model are x_t, $\hat{y}_t$, and $b_{\hat{y}_t}$, respectively.

3.2.3. Temporal fusion decoder

The temporal fusion decoder treats the encoder and decoder Bi-LSTM model construction output to enhance the local time information learning of past input and known future input data. The output of the Bi-LSTM model computes the GRN in advance to increase the depth of the model and avoid the gradient vanishing problem simultaneously. After that, the output of the GRNs generates queries Q, values V, and keys K vector for the masked interpretable multi-head attention training to learn the long-term time series relationships by its self-attention mechanism as below equation.

$$Attention_{dot-product}(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \tag{28}$$

where $\sqrt{d_k}$ represents the scaling factor to normalize the scale of dot-product attention output, the softmax activation function computes

the importance probability of each time stamp. Finally, the V vector combines with the time stamps importance probability to pay attention to essential time stamps.

However, in some complicated real-world problems, more than the depth of the model is needed to learn more information. The multi-head attention contaminates more than one attention mechanism output to enhance the model depth and learning capacity of time stamps see following equation.

$$MultiHead(Q,K,V) = [Head_1, Head_2, \ldots\ldots, Head_{h_{max}}]\, W_H, \tag{29}$$

$$Head_i = Attention_{dot-product}\left(QW_q^{(i)}, KW_k^{(i)}, VW_v^{(i)}\right), \tag{30}$$

where h_{max} represents the maximum attention heads number, W_H is the weight connect with different attention heads. The $W_q^{(i)}$, $W_k^{(i)}$, $W_v^{(i)}$ are the weight of Q, K, and V vectors, respectively, at number i head.

The interpretable multi-head attention improves the value sharing in each head by modifying the original self-attention head computation to the additive aggregation. This operation let the multi-head attention can interpret the features' importance because each head's weights are related by value-sharing calculation.

$$InterpretableMultiHead(Q,K,V) = \widetilde{Head}\, W_H, \tag{31}$$

$$\widetilde{Head}_i = \frac{1}{h_{max}} \sum_{i=1}^{h_{max}} Attention_{dot-product}\left(QW_q^{(i)}, KW_k^{(i)}, VW_v\right), \tag{32}$$

where $\widetilde{Head}_i$ is the number i head of interpretable multi-head attention. The W_H is the weight connect with different interpretable multi-head attention heads. The $W_q^{(i)}$, $W_k^{(i)}$ are the weight of Q, K, respectively, at number i head. The W_v is the sharing weights of V vector.

Finally, the outputs of interpretable multi-head attention are calculated by a GLU, LayerNorm, and a residual connection, and then the results to GRN. The above is the whole operation of the temporal fusion decoder.

3.2.4. Quantile output

The improved TFT model uses a linear transformation like TFT model to output temporal fusion decoder to generate the prediction intervals by quantile forecasts. The quantile prediction intervals simulate the predictions of various percentiles, such as 10 %, 50 %, and 90 %, at each prediction step.

$$\widehat{y}(q,t,\tau) = W_q \widetilde{\psi}(t,\tau) + b_q, \tag{33}$$

where W_q and b_q is the weight and bias of linear transformation layer for the specific quantile q. The $\widetilde{\psi}(t,\tau)$ represents the output of temporal fusion decoder at time $t+\tau$. The current time and future prediction step represent as t and, respectively. The $\widehat{y}(q,t,\tau)$ is the estimation of the improved TFT model for quantile q at prediction step $t+\ \tau$.

3.2.5. Loss function

Selecting a suitable loss function is essential for deep learning model training. It depends on the different problems you want to solve. The improved TFT model uses quintile loss to summate all quantile outputs as the original TFT model. See the equation below. To ensure the out-of-sample performance is accurate, the original TFT model measures the normalized quantile losses and considers and focuses on P50 and P90 risk.

$$L(\Omega, W) = \sum_{y_t \in \Omega} \sum_{q \in Q} \sum_{\tau=1}^{\tau_{max}} \frac{QL(y_t, \widehat{y}(q, t-\tau, \tau), q)}{M\tau_{max}}, \tag{34}$$

$$QL(y, \widehat{y}, q) = q(y - \widehat{y}) + (1-q)(\widehat{y} - y), \tag{35}$$

$$q - risk = \frac{2 \sum\limits_{y_{t \in \widetilde{\Omega}}} \sum\limits_{\tau=1}^{\tau_{max}} QL(y_t, \widehat{y}(q, t-\tau, \tau), q)}{\sum\limits_{y_{t \in \widetilde{\Omega}}} \sum\limits_{\tau=1}^{\tau_{max}} |y_t|} \tag{36}$$

where M represents the training samples of domain Ω, W is the improved TFT model weights, and Q is the set of output quintiles. The $\widetilde{\Omega}$ is the test samples domain. The t, τ, and τ_{max} represent the current time, future prediction step, and the maximum future prediction steps, respectively. The quantile loss (QL) is similar to the mean absolute error (MAE) for calculating the MAE of model output $\widehat{y}$ and actual y at quantile q.

4. Future RUL prediction results and discussion

The definition of the future RUL prediction assumes that the independent and dependent variables after the Sp are unknown. The definition of RUL is the cycle-life (hour) between the Sp of cycles and the number of cycles at the EOL as the following equation.

$$C_{T=future\ RUL} = C_{T=EOL} - C_{T=Sp}, \tag{37}$$

Where T and C represent the time unit (hour) and the cycle number, respectively.

The prediction model only can input these rare previous known variables and implement step-ahead rolling prediction to get the predicted values. The step-ahead rolling prediction in the following paragraph. The step-ahead rolling prediction conducts iteratively until the predicted value reaches the EOL threshold, then stop the prediction. Because of this stopping rule and the future RUL definition, the prediction and actual length would differ. For the reason of the length different between prediction and actual value, in this research the absolute error (AE) and relative errors (RE) use to be the performance matric, instead of mean squared error (MSE), mean absolute error (MAE), R-squared (R^2), etc. The formulation of performance matric of AE and RE define as below.

$$AE = |RUL_{actual} - \widehat{RUL}|, \tag{38}$$

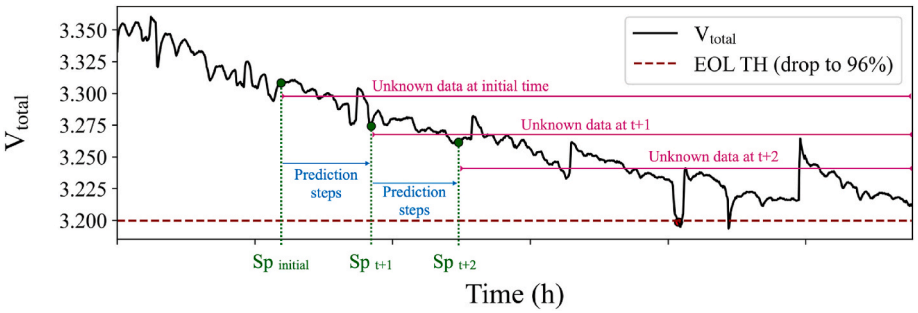


Fig. 9. Sp updating based on the step-ahead rolling RUL prediction for the FC1 dataset.

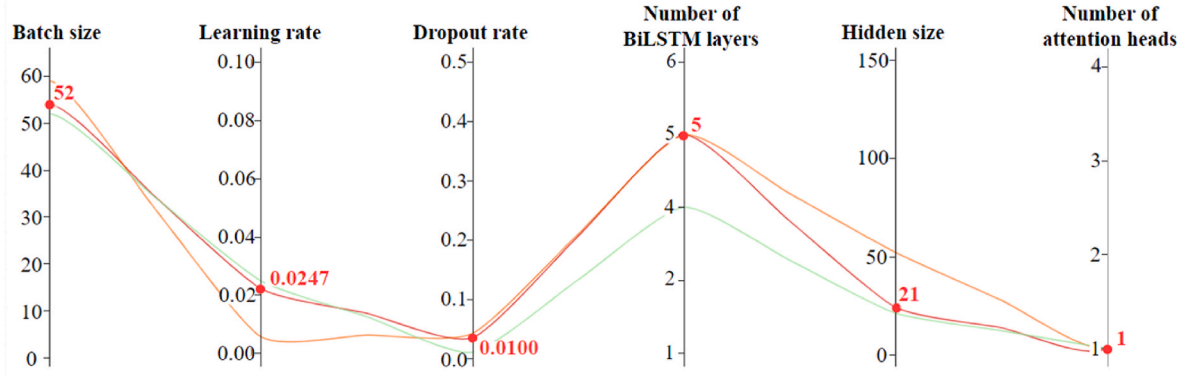


Fig. 10. The convergence process visualization of each hyperparameter for the top 1 % achieving better model by NNI toolkit for FC1 dataset.

Table 5
Hyperparameters optimization boundaries.

Hyperparameters	Type	Boundary
Batch size	Integer	[1, 64]
Learning rate	Float	[0.0001, 0.1]
Dropout rate	Float	[0.01, 0.5]
Number of BiLSTM layers	Integer	[1,6]
BiLSTM hidden size	Integer	[1, 150]
Number of attention heads	Integer	[2,4]

$$RE = \frac{|RUL_{actual} - \widehat{RUL}|}{RUL_{actual}}, \tag{39}$$

This study used three different EOL hours as three distinct scenarios for predicting future RUL. The first experiment involves an EOL of 815 h for the FC1 dataset, while the second and third experiments have an EOL of 313 and 878 h, respectively, for the FC2 dataset. The Sp calculated by the double Bacon Watts model for each experiment is 578 h for the FC1 dataset, 183 h for the FC2 dataset scenario#1, and 546 h for the FC2 dataset scenario#2. The first experiment involves an EOL of 815 h for the FC1 dataset, while the second and third experiments have EOLs of 313 and 878 h, respectively, for the FC2 dataset. The actual RULs for the FC1 dataset, FC2 dataset scenario#1, and FC2 dataset scenario #2 are 237 h, 210 h, and 277 h, respectively. The selection of features used for model training can enhance model performance and reduce computing consumption. The selection of features used for model training can enhance model performance and reduce computing consumption, as shown in Table 4 for each experimental scenario.

Implementing step-ahead rolling prediction for future RUL prediction enables the use of the improved TFT model to predict single or multiple steps in FC1 and FC2 scenarios. The prediction step starts from the Sp to predict the next single or multiple steps. The first Sp is called initial prediction Sp. After getting the next single or multiple steps, the Sp will update based on the prediction step, and over time, the unknown data will update based on the Sp updating. To simply assess the performance of the proposed improved TFT model in predicting the next 1-h, single step-ahead rolling prediction is utilized for FC1, FC2_scenario#1, and FC2_scenario#2, as depicted in Fig. 9.

During model training, the original training set, which includes data points from knee-onset to knee-point, is divided into a sub-training set and a validation set. The sub-training and validation sets are 90 % and 10 % of the original training set, respectively. This division is crucial for tuning the hyperparameters of the proposed TFT model. The hyperparameters are adjusted based on the model performance by calculating the objective function from the validation set during optimization. This research uses the Neural Network Intelligence (NNI) open-source AutoML toolkit to optimize hyperparameters. This AutoML toolkit provides a web-based graphical user interface (GUI) that facilitates a better understanding of each optimization iteration's hyperparameter optimization convergence process. This research selects the BO-based TPE algorithm to optimize the hyperparameters due to its outstanding performance. The search space of the TPE algorithm included some parameters worse and better group condition based on the top-quantile ratio [23]. The kernel density estimators (KDEs) of TPE algorithm are given by

$$p(x|y,D) = \begin{cases} p(x|y,D^{(l)}), (y \le y^{\gamma}) \\ p(x|y,D^{(g)}), (y \rangle y^{\gamma}) \end{cases} \tag{40}$$

where x represents the hyperparameters, y represents the output of the objective function, D represents the set of observations, and γ represents the top quantile for the better groups. l and g are used to denote the better and worse groups, respectively.

The convergence process of each hyperparameter for the top 1 % achieving better model performance for the FC1 dataset on the validation set is illustrated in Fig. 10. The red color connecting lines with hyperparameters represent the combinations that yield the most accurate predictions on the validation set and the other colors represent the six hyperparameter combinations with better accurate predictions. The boundaries for each hyperparameter are outlined in Table 5. In 300 iterations of BO-based TPE optimization, the optimal hyperparameters for each scenario: for the FC1 dataset experiment, batch size = 52, learning rate = 0.0247, dropout rate = 0.0100, number of BiLSTM layers = 4, hidden size = 21, and number of attention heads = 1; for FC2_scenario

Table 6
Future RUL prediction results for the three experiments.

Dataset	Actual RUL	Predicted RUL	Predicted RUL interval	AE	RE (%)
FC1	237	237.1749	[237.0773, 237.3510]	0.1749	0.0738
FC2_scenario#1	208	210.1297	[210.1296, 211.0107]	2.1297	1.0239
FC2_scenario#2	112	110.1507	[110.0113, 111.0074]	1.8493	1.6512

Table 7
Future RUL prediction results without feature selection for the three experiments.

Dataset	Actual RUL	Predicted RUL	Predicted RUL interval	AE	RE (%)
FC1	237	235.0069	[235.0068, 235.2339]	1.9931	0.8410
FC2_scenario#1	208	212.4118	[208.2253, 212.4118]	4.4118	2.1210
FC2_scenario#2	112	109.0087	[109.2888, 110.1501]	2.9913	2.6708

#1, batch size = 40, learning rate = 0.0756, dropout rate = 0.1437, number of BiLSTM layers = 6, hidden size = 14, and number of attention heads = 4; and similarly, for FC2_scenario#2, the optimal hyperparameters are batch size = 50, learning rate = 0.0263, dropout rate = 0 0.29105, number of BiLSTM layers = 1, and hidden size = 30, and number of attention heads = 4.

Table 6 shows the online RUL prediction with single prediction step results using optimal hyperparameters for the three experiments. Based on the predicted results in Table 6, the proposed approach demonstrates high accuracy and narrow predicted RUL intervals across the three scenarios, indicating its reliability and low uncertainty. The prediction interval ranges are 0.2737, 0.8811, and 0.9961 for FC1, FC2_scenario#1, and FC2_scenario#2, respectively. In addition, the online RUL prediction with a single prediction step without feature selection is also shown in Table 7. From Table 7, the future RUL prediction results without feature selection show lower accuracy than those obtained with the proposed feature selection method, along with more extensive prediction interval ranges. The most significant impact is observed in FC2_scenario#1, which can be attributed to the strong influence of appropriate feature selection on prediction accuracy, particularly given the limited length of training data. Feature selection helps the deep learning model focus on the most relevant features, reducing training time, prediction time consumption, and uncertainty in the results. Fig. 11 presents the online RUL prediction with single prediction step results with feature selection for Table 6. The results show that the proposed model outperforms each experiment, and the rolling prediction results are very close to the actual unknown future V_{total}. Although the training length is concise and there are various data trends, such as the FC2 scenario#1 and #2 experiments, the proposed model can predict future V_{total} trends accurately.

In this study, multiple online RUL prediction steps are analyzed to evaluate and compare the prediction horizon of the proposed improved TFT model. This assessment aims to gauge the model's capability to deliver precise predictions for future V_{total}. Achieving accuracy in a significant prediction step allows for early detection of the future trend of V_{total}, thereby preventing sharp battery degradation. The analysis focuses on the impact of prediction steps 1, 3, 5, 7, and 9. The online prediction results of AE and RE for the improved TFT model in different prediction steps are shown in Table 8. Fig. 12 presents the online RUL prediction results with 12 different prediction steps for Table 8. In Tables 8 and it shows that the prediction steps significantly affect the online RUL prediction performance for FC2 dataset scenario#1. With only 55 data points for the training length in FC2 scenario#1, accurate predictions for more significant steps are challenging for the model. The analysis of various prediction steps in Table 8 shows that the improved TFT model achieves accurate predictions for five steps. Despite the limited training data points, this accuracy is consistently maintained across all experiments, including FC2 dataset scenario#1. The AE in each experiment remain below 10, affirming the model's capability to predict the next 5-h V_{total} precisely. The proposed model with the proposed modeling procedure gets accurate results for each prediction step from one to 9 h. Moreover, from Table 8, increasing the prediction step causes the proposed model to get worse RUL prediction results, which can double verify the research procedure is reasonable because the more predicted steps produce more prediction bias. Ultimately, the effectiveness of a DL model, considering both training and testing speeds, is crucial and directly correlates with the intrinsic value of the utilized model. Table 9 illustrates the computational time for training and prediction of the proposed model across three scenarios. As depicted in Table 9, the proposed improved TFT model exhibits high-speed performance in training and prediction processes.

Due to the scarcity of open-source PEMFC datasets, it is challenging to locate an alternative dataset for testing the model's performance.

Table 8
Online RUL prediction results with 12 different prediction steps for three experiments.

Dataset	Steps	Actual RUL	Predicted RUL	AE	RE (%)
FC1	1	237	237.1749	0.1749	0.0738
	3		236.0726	0.9274	0.3913
	5		238.1313	1.1313	0.4774
	7		241.9920	4.9920	2.1063
	9		242.6911	5.6912	2.4013
FC2_scenario#1	1	208	210.1297	2.1297	1.0239
	3		212.4117	4.4118	2.1210
	5		214.1400	6.1401	2.9519
	7		223.8890	15.8890	7.6389
	9		224.8706	16.8706	8.1108
FC2_scenario#2	1	112	110.1507	1.8493	1.6512
	3		119.1868	7.1868	6.4168
	5		119.8686	7.8685	7.0255
	7		118.1099	6.1099	5.4552
	9		125.0479	13.0479	11.6499

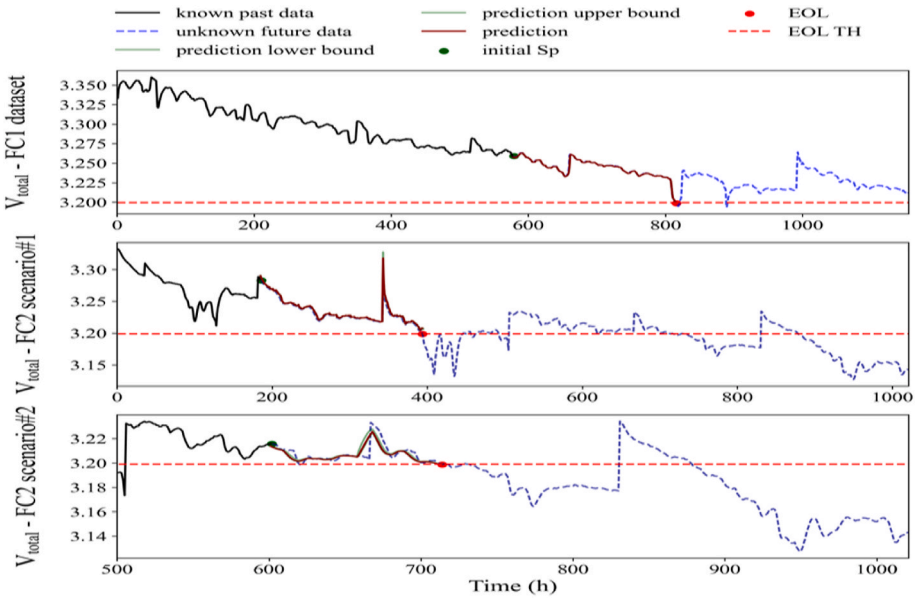


Fig. 11. The future RUL prediction results for the three experiments with one prediction step.

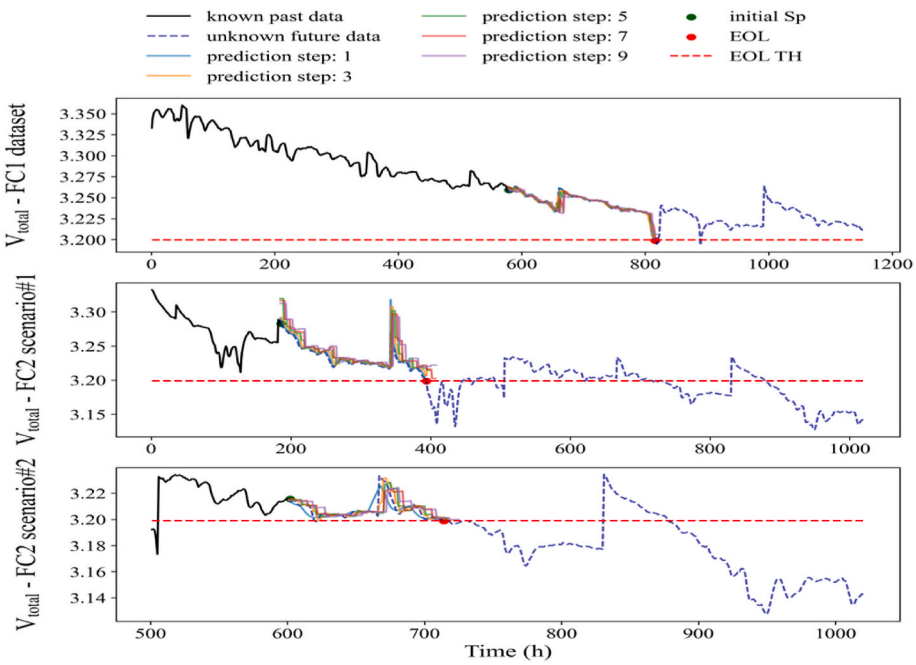


Fig. 12. The online RUL prediction results with 12 different prediction steps for three experiments.

Table 9
Proposed model on training time and prediction time.

Dataset	Training length	Training time	Prediction time (Average of each prediction step)
FC1	116–578	32.9533 s	0.5653 s
FC2_scenario#1	147–186	10.9496 s	0.5699 s
FC2_scenario#2	568–609	15.0362 s	0.5585 s

Table 10
Comparison results with existed research for the FC1 experiment.

Dataset	EOL	Training length	Model	Actual RUL	Predicted RUL	RE (%)
FC1	812 h	300 h	LSTM [8]	512 h	513.7600 h	0.3437
			BiLSTM [8]	[8]	511.6700 h	0.0645
			ESN [8]		518.3800 h	1.2461
			TFT		512.1081 h	0.0211
			Improved TFT		**512.0760 h**	**0.0148**
		400 h	LSTM [8]	412 h	414.3900 h	0.5801
			BiLSTM [8]	[8]	411.4200 h	0.1408
			ESN [8]		410.5600 h	0.3495
			TFT		412.9673 h	0.2348
			Improved TFT		**412.0248 h**	**0.0060**
		500 h	LSTM [8]	312 h	312.1400 h	0.0449
			BiLSTM [8]	[8]	311.2400 h	0.2436
			ESN [8]		310.6800 h	0.4231
			TFT		311.1098 h	0.2853
			Improved TFT		**312.2067 h**	**0.0662**
		600 h	LSTM [8]	212 h	211.2500 h	0.3538
			BiLSTM [8]	[8]	211.3000 h	0.3301
			ESN [8]		210.9200 h	0.5094
			TFT		212.1076 h	0.0508
			Improved TFT		**212.1029h**	**0.0486**

Table 11
Comparison results of different models for the FC2 dataset.

Dataset	EOL	Training length	Model	Actual RUL	Predicted RUL	RE (%)
FC2	395 h	300 h	LSTM [8]	95 h	94.4800 h	0.5474
			BiLSTM [8]	[8]	94.8500 h	0.1579
			ESN [8]		94.6700 h	0.3474
			TFT		94.2226 h	0.8270
			Improved TFT		**95.0307 h**	**0.0326**

Consequently, this research opted to compare the experiment results with existing research [8] and conduct additional experiments to assess the proposed method's robustness and performance. In the compared experiment, the same training length, features selected by random forest, and optimal hyperparameters are employed to ensure a fair comparison. This comparison uses the same training length to compare the proposed method: high peak removal, data smoothing, knee-onset, feature selection by RF model, improved TFT model, hyperparameters tuning, and step-ahead prediction. Because the Sp should be the same as the existing research [8], the knee-point estimation cannot be used for the comparison only for the three scenarios. For FC1 and FC2, refer to the results in Tables 10 and 11. The compared existing research [8] uses EOL thresholds 3.211 V and 3.197 V for the FC1 and FC2 datasets, which is the original consideration. Because the data preprocessing approach involves merging the original data into hourly data differences, it can result in variations in the actual EOL and RUL values. The EOL thresholds 3.211 V and 3.197 V for the FC1 and FC2 datasets used in this research cannot get the same EOL number as the existing research [8]. Thus, in this comparison, the results of the FC1 and FC2 experiments directly use the actual EOL cycle number as the same number of 812 and 395 as the existing research [8]. The two EOL cycle numbers 812 and 395 can be used to compare directly because these two EOL cycle numbers are in the continuous degradation trend. The EOL cycle number of 812 and 395 matches to our FC1 and FC2 data is at EOL threshold in value of 3.203 V and 3.183 V. Tables 10 and 11 show the RE value to compare the performance of the proposed method and the existing research [8]. From Tables 10 and 11, the proposed method

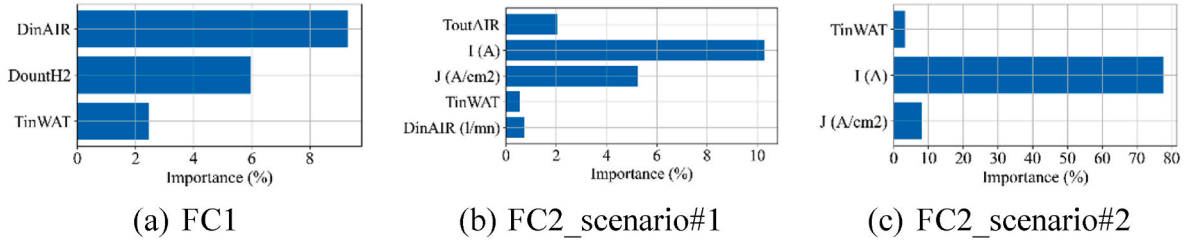


Fig. 13. Variable importance is calculated using the proposed improved TFT model.

outperformed the existing research [8] and the RE is 0.0148 %, 0.0060 %, 0.0662 %, and 0.0486 % for the FC1 dataset, and 0.0326 % for the FC2 dataset. The predicted RUL in these five experiments is very close to the actual.

The most important feature of the TFT model is its ability to provide excellent interpretability, enabling the identification of contributions for the input features. The following presents the variable contribution, calculated using the improved TFT model proposed in Fig. 13, for our specific scenarios. The rankings of the contribution to enhancing the proposed improved TFT model performance in our specific scenarios are as follows:

- FC1 is DinAIR > DoutH2 > TinWAT.
- FC2_ scenario1 is I > J > TinAIR > DinAIR > TinWAT.
- FC2 _scenario_2 is I > J > TinWAT.

Drawing insights from this model's interpretability, the contribution of each feature to the proposed model becomes apparent. When too many features are input, the model requires significant training consumption. The contribution results can serve as a guide to selectively remove input features, thereby reducing training time.

5. Conclusions

Predicting the online RUL of the PEMFC pack is challenging, especially under dynamic operation conditions, but the proposed approach with improved TFT model can tackle it. This paper analyzes explicitly the PEMFC dataset and predicts the health indicator of V_{toal} through three analysis parts: raw data analysis, modeling, and future RUL prediction. In raw data analysis, the suitable interpolation approach is used to remove outliers and knee-onset and knee-point identification to determine the training set for the proposed model. The RF model selects essential features at the end of raw data analysis. This research uses PCA's feature domain analysis to verify and visualize that our raw data analysis is reasonable and makes sense. For modeling, the determined training set is used to train the proposed improved TFT model, which changes the encoder module from LSTM to BiLSTM. During model training, this research combines with the NNI AutoML toolkit to optimize the hyperparameters. The toolkit provides hyperparameter converge process visualization, which can easily understand the optimization process. After the proposed model training, the train model with optimal hyperparameters is used for online RUL prediction.

This study builds one scenario for the FC1 dataset and two scenarios for the FC2 dataset by determining the different EOL by ourselves, four different training lengths for the FC1 dataset, and one for the FC2 dataset to compare the model performance with other existing research. The training length determination of comparison follows the existing research. The online RUL results of the proposed improved TFT model demonstrate under static (FC1 dataset) conditions of the comparison with other research shows that the RE of the online RUL prediction performance of the improved TFT model for training data [0,300 h], [0,400 h], [0,500 h], and [0,600 h] are 0.0148 %, 0.0060 %, 0.0662 %, and 0.0486 %, respectively. Under dynamic operation (FC2 dataset) conditions of the comparison with other research, the RE of the online RUL prediction performance of the improved TFT model for training data [0,300 h] is 0.0326 %. The experimental results demonstrate that the RE is lower than 0.07 % at different training durations and has better prediction performance than ESN, LSTM, and BiLSTM. In our build experiment, the online RUL prediction RE values for the three experiments are 0.0738 %, 1.0239 %, and 1.6512 %, respectively.

The most important finding in this study is that we tested five different multi-step-ahead parameters to determine the prediction capability limits of the proposed improved TFT model. Through these experiments, we found that the improved TFT model can provide reliable five-step-ahead predictions with minimal accuracy loss and low prediction uncertainty, as verified through testing across various prediction steps. Additionally, this study employs PCA to visualize the domain differences between models with and without feature selection, highlighting the importance of feature selection in deep learning model training. Our proposed approach's analysis procedure integrates multiple methodological components, which collectively enhance the future prediction performance of the improved TFT model. The outcomes of each experiment highlight that the proposed approach, designed for robustness and high reliability in online RUL prediction, can effectively address real-world PEMFC future prediction challenges. In the future, we will explore the application of the improved TFT model in transfer learning for PEMFC lifetime prediction. Due to the scarcity of battery degradation data, transfer learning can adapt model weights from source battery data to target battery data, enabling reliable predictions with limited target data. We will focus on domain adaptation, a widely used transfer learning technique that mitigates domain shifts between source and target datasets. Specifically, we aim to address temporal feature disparities and optimize model weights to align with the target domain. This approach will enhance future RUL predictions while reducing the data collection effort for researchers.

CRediT authorship contribution statement

Jia-Hong Chou: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation. **Fu-Kwun Wang:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Science and Technology Council, Taiwan under Grant NSTC-112-2221-E011-114-MY3.

References

[1] W. Zhu, B. Guo, Y. Li, Y. Yang, et al., Uncertainty quantification of proton-exchange-membrane fuel cells degradation prediction based on bayesian-gated recurrent unit, eTransportation 16 (2023) 100230.

[2] J. Zuo, H. Lv, D. Zhou, Q. Xue, et al., Deep learning based prognostic framework towards proton exchange membrane fuel cell for automotive application, Appl. Energy 281 (2021) 115937.
[3] R. Ma, T. Yang, E. Breaz, et al., Data-driven proton exchange membrane fuel cell degradation predication through deep learning method, Appl. Energy 231 (2018) 102–115.
[4] Z. Hua, Z. Zheng, E. Pahon, et al., Remaining useful life prediction of PEMFC systems under. dynamic operating conditions, Energy Convers. Manag. 231 (2021) 113825.
[5] M. Yue, S. Jemei, N. Zerhouni, R. Gouriveau, Proton exchange membrane fuel cell system prognostics and decision- making: current status and perspectives, J Renew Energy 179 (2021) 2277–2294.
[6] C. Zhang, Y. Zhang, L. Wang, et al., A health management review of proton exchange membrane fuel cell for electric vehicles: failure mechanisms, diagnosis techniques and mitigation measures, Renew. Sustain. Energy Rev. 182 (2023) 113369.
[7] T. Zihan, Z. Chu, X. Jinlin, et al., Evolutionary gate recurrent unit coupling convolutional neural network and improved manta ray foraging optimization algorithm for performance degradation prediction of PEMFC, Appl. Energy 336 (2023) 120821.
[8] H. Li, Q. Chen, L. Zhang, et al., Degradation prediction of proton exchange membrane fuel cell based on the multi-inputs bi-directional long short-term memory, Appl. Energy 344 (2023) 121294.
[9] X. Sun, M. Xie, J. Fu, et al., An improved neural network model for predicting the remaining useful life of proton exchange membrane fuel cells, Int J Hydrog Energy 48 (65) (2023) 25499–25511.
[10] R. Xie, C. Li, R. Ma, et al., An explainable data-driven framework for fuel cell aging prediction under dynamic condition, IEEE Trans. Ind. Electron. 70 (6) (2023) 5960–5970.
[11] B. Sun, X. Liu, J. Wang, et al., Short-term performance degradation prediction of a commercial vehicle fuel cell system based on CNN and LSTM hybrid neural network, Int J Hydrog Energy 51 (2) (2023) 18–25.
[12] Z. Deng, S.H. Chan, Q. Chen, et al., Efficient degradation prediction of PEMFCs using ELM-AE based on fuzzy extension broad learning system, Appl. Energy 331 (2023) 120385.
[13] Y. Peng, T. Chen, F. Xiao, S. Zhang, Remaining useful lifetime prediction methods of proton exchange membrane fuel cell based on convolutional neural network-long short-term memory and convolutional neural network-bidirectional long short-term memory, Fuel Cells 23 (1) (2023) 75–87.
[14] J. Jin, Y. Chen, C. Xie, F. Wu, et al., Remaining useful life prediction of PEMFC based on the multi-input cycle reservoir with jump network, Int J Hydrog Energy 48 (2023) 12844–12860.
[15] K. Benaggoune, M. Yue, S. Jemei, N. Zerhouni, A data-driven method for multi-step-ahead prediction and long-term prognostics of proton exchange membrane fuel cell, Appl. Energy 313 (2022) 118835.
[16] V. Bilgram, F. Laarmann, Accelerating innovation with generative AI: AI-augmented digital prototyping and innovation methods, IEEE Eng. Manag. Rev. 48 (2023) 12844–12860.
[17] B. Lim, S.Ö. Arık, N. Loeff, et al., Temporal fusion transformers for interpretable multi-horizon time series forecasting, Int. J. Forecast. 37 (2021) 1748–1764.
[18] P. Fermín-Cueto, E. McTurk, M. Allerhand, et al., Identification and machine learning prediction of knee-point and knee-onset in capacity degradation curves of lithium-ion cells, Energy AI 1 (2020) 100006.
[19] Mao L, Jackson L. IEEE 2014 Data Challenge Data. Loughborough University. Dataset. https://doi.org/10.17028/rd.lboro.3518141.v1.
[20] S. Li, W. Luan, C. Wang, Y. Chen, Z. Zhaung, Degradation prediction of proton exchange membrane fuel cell based on Bi-LSTM-GRU and ESN fusion prognostic framework, Int J Hydrog Energy 78 (12) (2022) 33466–33478.
[21] Z. Hua, Z. Zheng, S.C. Péra, F. Gao, Remaining useful life prediction of PEMFC systems based on the multi-input echo state network, Appl. Energy 265 (2020) 114791.
[22] Z. Zhang, Y.X. Wang, H. He, F. Sun, A short- and long-term prognostic associating with remaining useful life estimation for proton exchange membrane fuel cell, Appl. Energy 304 (2021) 117841.
[23] S. Watanabe, Tree-structured parzen estimator: understanding its algorithm components and their roles for better empirical performance [Online]. Available: https://doi.org/10.48550/arXiv.2304.11127, 2023.